

Recall from last class:

- For a matrix A , define A_{ij} to be the matrix obtained from A by deleting row i and column j .
- For $n \geq 2$, the determinant of an $n \times n$ matrix $A = [a_{ij}]$ is

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + \cdots + (-1)^{n+1} a_{1n} \det(A_{1n}) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}).$$

We call this the cofactor expansion across the first row.

- **Mohan's Theorem:** Cofactor expansion across any row or column is the same, and hence each cofactor expansion gives the determinant.
 - Let $A = [a_{ij}]$ be an $n \times n$ matrix, then A is **upper triangular** if $a_{ij} = 0$ when $i > j$.
- Theorem 3.2:** Let A be an $n \times n$ upper triangular matrix, then

$$\det(A) = a_{11} a_{22} \cdots a_{nn} = \prod_{j=1}^n a_{jj}.$$

- **Theorem 3.3:** If A is an $n \times n$ matrix and E is an elementary matrix, then

$$\det(EA) = \det(E) \det(A),$$

and

$$\det(E) = \begin{cases} 1 & \text{if } E \text{ represents adding a multiple of a row to another row,} \\ -1 & \text{if } E \text{ represents interchanging two rows,} \\ r & \text{if } E \text{ represents scaling a row by a factor of } r. \end{cases}$$

Theorem 3.6: If A and B are n by n matrices, then

$$\det(AB) = \det(A) \det(B).$$

1. Prove the theorem.

2. **Corollary:** A is invertible if and only if $\det(A) \neq 0$. Prove the corollary.

3. True or False:

(a) If A is an $n \times n$ invertible matrix, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

(b) If A is an $n \times n$ matrix, then $\det(A) = \det(A^T)$.

(c) If $A \sim B$, then $\det(A) = \det(B)$.

4. We can think of $\det$ as a function from the set of $n \times n$ matrices to the real numbers. Is this function linear?

5. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(\vec{x}) = A\vec{x}$ where $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Let S_1 be the unit square given by $S_1 = \{(x, y) : 0 \leq x, y \leq 1\}$.

(a) Find $T(S_1)$.

(b) How does the area of S_1 related to the area of $T(S_1)$?

(c) Suppose $S_2 = \{(x, y) : 0 \leq x, y \leq 2\}$. How does the area of S_2 related to the area of $T(S_2)$?

6. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(\vec{x}) = A\vec{x}$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Let S be the unit square given by $S = \{(x, y) : 0 \leq x, y \leq 1\}$.

(a) Find $T(S)$.

(b) How does the area of S related to the area of $T(S)$?

Theorem: If A is a 2×2 matrix, then the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, then the volume of the parallelepiped is $|\det A|$.

7. Find the area of the parallelogram with vertices $(-1, 0)$, $(0, 5)$, $(1, -4)$, and $(2, 1)$.

Theorem: Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be linear with associated matrix A . If S is a parallelogram in $\mathbb{R}^2$, then $\text{Area}(T(S)) = |\det A| \text{Area}(S)$.

We will generalize this theorem in Math 22b using calculus to more general functions!